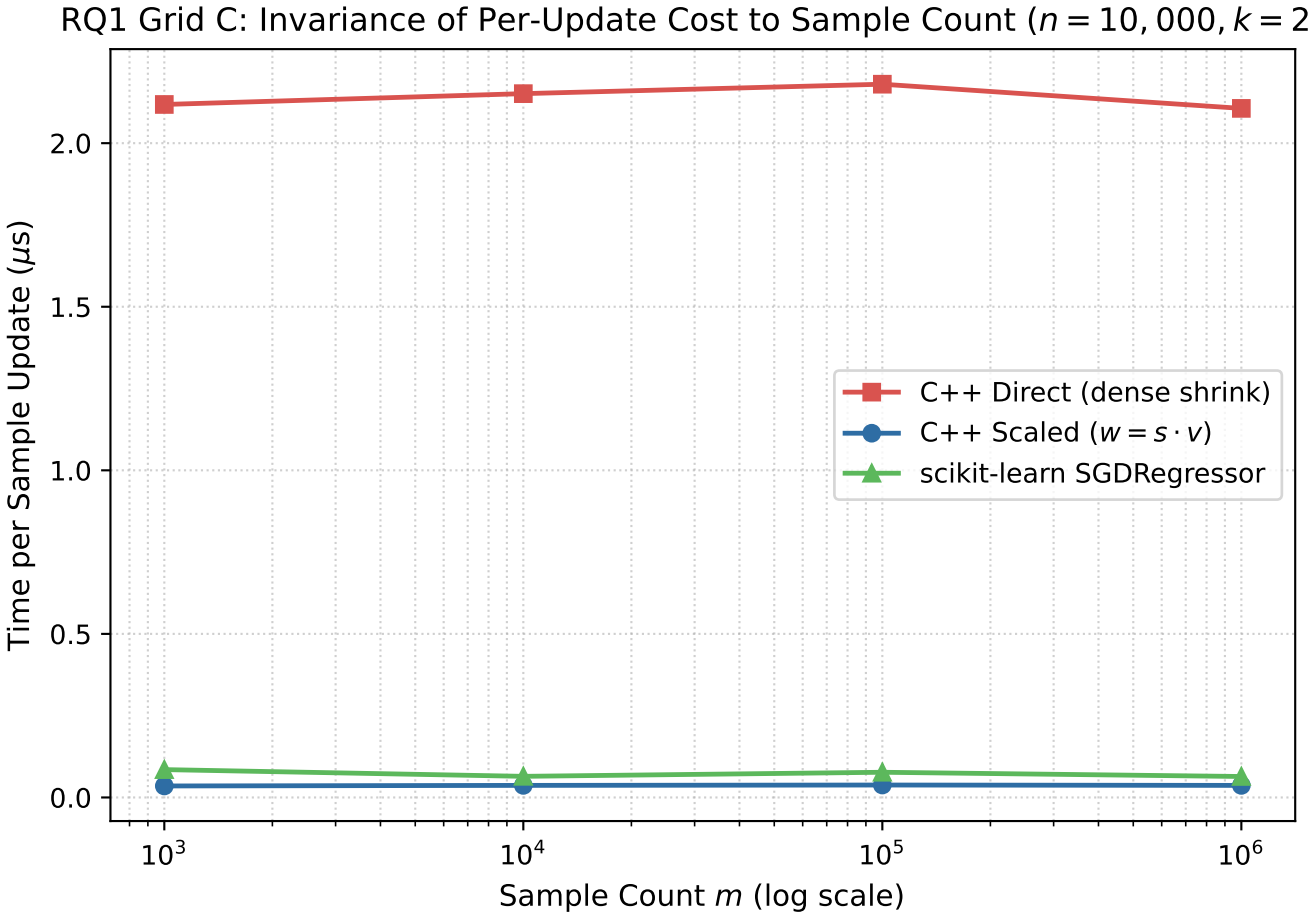
RQ1 Grid C: Invariance of Per-Update Cost to Sample Count ($n = 10,000, k = 20$
Time per Sample Update (μs)
2.0
1.5
1.0
0.5
0.0
10^3
10^4
10^5
10^6
Sample Count m (log scale)
C++ Direct (dense shrink)
C++ Scaled ($w = s \cdot v$)
scikit-learn SGDRegressor